Yong Guk Kang

Research Professor, Imaging Intelligence Laboratory, School of Mechanical Engineering, Seoul National University
ygkang@snu.ac.kr | ORCID: 0000-0002-5572-7544

SUMMARY

Research Professor at Seoul National University's Imaging Intelligence Laboratory, working on computational imaging, inverse optical design, lensless imaging, quantitative phase imaging, and biomedical photonics. My research connects optical system modeling, wave-optics simulation, and data-driven reconstruction for imaging problems where hardware design and computational inference must be considered together.

CURRENT RESEARCH INTERESTS

- Computational imaging and inverse optical design
- Lensless and phase-coded imaging systems
- AI-assisted image reconstruction and virtual staining
- Quantitative phase, interferometric, and coherent imaging
- Physics-informed modeling of optical imaging systems

PREVIOUS RESEARCH THEMES

- Optical coherence microscopy for cell and tissue dynamics
- Nonlinear microscopy for cell-ECM mechanical interactions
- Biomedical photonics and quantitative bioimage analysis
- Fluorescence optical mapping and multimodal microscopy

EDUCATION

Ph.D. in Biomedical Engineering, Korea University (2014-03-2020-08)
Thesis: Development of a multiphoton and optical coherence microscope for visualization and analysis of cell-extracellular matrix interactions

- Implemented live-cell multiphoton and optical coherence microscopy for cell and tissue imaging.
- Analyzed 3D matrix deformation using digital volume correlation.

B.S. in Biomedical Engineering, Korea University (2008-03-2014-02)

POSITIONS

Research Professor, Seoul National University, School of Mechanical Engineering (2024-04-present)

- Developing computational imaging frameworks that combine optical system modeling, inverse design, and data-driven reconstruction.
- Investigating lensless and phase-coded imaging strategies for high-dynamic-range and information-rich optical measurements.
- Applying physics-informed simulation and machine learning to jointly reason about optical encoding and image reconstruction.

Postdoctoral Researcher, Korea University, School of Biomedical Engineering (2022-09-2024-03)

- Advanced reflection phase microscopy for efficient 3D quantitative imaging with reduced axial scanning.
- Developed computational refocusing and field synthesis strategies for reflection phase measurements.
- Integrated optical modeling and experimental validation for interferometric biomedical imaging.

Research Lecturer, Korea University, School of Medicine (2020-10-2022-08)

- Developed optical coherence microscopy approaches for quantifying cell-ECM interactions and intracellular dynamics.
- Applied temporal-dynamics contrast imaging to live-cell and tissue-scale biomedical measurements.
- Connected optical imaging measurements with quantitative analysis of mechanical interactions in biological matrices.

SELECTED CAPABILITIES

Computational imaging: Wave-optics simulation, Inverse problems and image reconstruction, Machine learning assisted optical design and reconstruction

Optical imaging: Quantitative phase imaging, Interferometric microscopy, Optical coherence microscopy, Nonlinear microscopy

Tools: MATLAB, Python, PyTorch, ZEMAX, Fusion 360, ImageJ

SELECTED PUBLICATIONS

Hyesuk Chae and Kyung Chul Lee and Yong Guk Kang and Jongho Kim and Kyungwon Lee and Suki Kang and Geunbae Bang and Nam Hoon Cho and Seung Ah Lee. **High-speed virtual re-staining via near-infrared quantitative phase imaging**. *Proceedings of SPIE Advanced Biophotonics Conference*, 2026.

Kwanjun Park and Taeseok Daniel Yang and Yong Guk Kang and Taeyoon Kong and Youngwoon Choi. **Wide-area topography using reflection phase microscopy for surface inspection**. *Optics Express*, 2025.

Kyung Chul Lee and Hyesuk Chae and Jongho Kim and Lucas Kreiss and Hyeongyu Kim and Yong Guk Kang and Kevin C. Zhou and Amey Chaware and Kanghyun Kim and Shiqi Xu and Suki Kang and Geunbae Bang and Nam Hoon Cho and Dosik Hwang and Roarke Horstmeyer and Seung Ah Lee. **Semi-supervised virtual staining using learned-illumination Fourier ptychography for high-speed label-free histopathology**. *Journal of Physics: Photonics*, 2025.

Yong Guk Kang and Donggeon Bae and Nakkyu Baek and Hyojun Ahn and Taeyoung Kim and Kyung Chul Lee and Seung Ah Lee. **Non-Paraxial Layered Diffraction Simulator for Inverse Design of Lensless Imaging Optics**. *Optica Imaging Congress 2025*, 2025.

Yongjun Jang and Myeongjin Kang and Yong Guk Kang and Dongtak Lee and Hyo Gi Jung and Dae Sung Yoon and Jongseong Kim and Yongdoo Park. **Cardiac fibroblast-mediated ECM remodeling regulates maturation in an in vitro 3D engineered cardiac tissue**. *Journal of Tissue Engineering*, 2025.

Hyun-Joong Kim and Se-Hwan Um and Yong Guk Kang and Myeongjin Shin and Hyeon Jeon and Beop-Min Kim and Dongtak Lee and Kisoo Yoon. **Laser-tissue interaction simulation considering skin-specific data to predict photothermal damage lesions during laser irradiation**. *Journal of Computational Design and Engineering*, 2023.

Yong Guk Kang and Kwanjun Park and Min Gu Hyeon and Taeseok Daniel Yang and Youngwoon Choi. **Three-dimensional imaging in reflection phase microscopy with minimal axial scanning**. *Optics Express*, 2023.

Yong Guk Kang and R. J. E. Canoy and Yujin Jang and A. R. M. P. Santos and Inseong Son and Beop-Min Kim and Yongdoo Park. **Optical coherence microscopy with a split-spectrum image reconstruction method for temporal-dynamics contrast-based imaging of intracellular motility**. *Biomedical Optics Express*, 2023.

Yong Guk Kang and Hwanseok Jang and Yongdoo Park and Beop-Min Kim. **Development of a 3-D Physical Dynamics Monitoring System Using OCM with DVC for Quantification of Sprouting Endothelial Cells Interacting with a Collagen Matrix**. *Materials*, 2020.

Min Gu Hyeon and Taeseok Daniel Yang and Jong Su Park and Kwanjun Park and Yong Guk Kang and Beop-Min Kim and Youngwoon Choi. **Reflection Phase Microscopy by Successive Accumulation of Interferograms**. *ACS Photonics*, 2019.

Yong Guk Kang and Hwanseok Jang and Taeseok Daniel Yang and Jacob Notbohm and Youngwoon Choi and Yongdoo Park and Beop-Min Kim. **Quantification of focal adhesion dynamics of cell movement based on cell-induced collagen matrix deformation using second-harmonic generation microscopy**. *Journal of Biomedical Optics*, 2018.

Taeseok Daniel Yang and Kwanjun Park and Yong Guk Kang and Kyoung J. Lee and Beop-Min Kim and Youngwoon Choi. **Single-shot digital holographic microscopy for quantifying a spatially-resolved Jones matrix of biological specimens**. *Optics Express*, 2016.

PROJECTS

Lensless and HDR computational imaging. Computational imaging research on lensless and phase-coded acquisition, with emphasis on optical encoding, inverse design, and robust reconstruction under challenging measurement conditions.

Reflection phase microscopy. Quantitative reflection phase microscopy for efficient 3D surface and biomedical imaging, combining interferometric measurement design with computational refocusing and field synthesis.

Optical coherence microscopy for cell-ECM dynamics. Optical coherence microscopy and temporal-dynamics contrast imaging for studying cell, tissue, and extracellular matrix interactions.

Live-cell nonlinear microscopy for matrix deformation analysis. Nonlinear microscopy and quantitative image analysis for studying collagen matrix deformation and cell-ECM mechanical interactions.

FUNDING, FELLOWSHIPS, AND PATENTS

Funding: Creative and Challenging Research Base Support, National Research Foundation of Korea (2021R1I1A1A01048370)

Fellowship: Global Ph.D. Fellowship, National Research Foundation of Korea (2015H1A2A1030048)

Patent: Head mount system for providing surgery support image, United States Patent (US 11,356,651)

Patent: Reflection phase microscopy system for 3D image acquisition, Korea Patent (10-2024-0081557)

Patent: Phase mask fabrication method based on grayscale lithography, Korea Patent (10-2024-0161144)